An Exact Cooperation Formula for Introspection Dynamics in the Heterogeneous Public Goods Game

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Abstract

Cooperation in heterogeneous groups, where individuals differ in resources, productivity, and behavioural responsiveness, underpins collective action across many social and biological systems. Introspection dynamics, in which each player compares their payoff to what they would have received under the alternative action, provides a natural learning rule for such asymmetric settings. We study introspection dynamics on multiplayer games in which the payoff difference Δf_i evaluated by a player when considering a strategy switch is independent of all other players' current actions, a property we call *state-independence*. Under this condition the introspection Markov chain decomposes as a random-scan product of N independent two-state chains, one per player, and the stationary distribution is a product measure. As our main application we consider the heterogeneous public goods game, where N players may differ in their contributions α_i, public goods multipliers r_i, and selection intensities β_i. We prove that the linear payoff structure implies state-independence, and the long-run cooperation probability admits the exact closed form

$$p_C^{\text{intro}} = \frac{1}{N} \sum_{i=1}^{N} \left[\frac{1 - \mu_{i0} - \mu_{i1}}{1 + e^{\beta_i \alpha_i (1 - r_i/N)}} + \mu_{i0} \right],$$

with no asymptotic approximation. Several structural consequences follow immediately: a player-specific cooperation threshold at $r_i = N$ (under symmetric mutation), payoff-neutrality under zero selection intensity, and the sign of each player's sensitivity to their own parameters.

1 Introduction

The public goods game is a canonical model of cooperation in evolutionary game theory: players decide whether to contribute to a common resource, which is multiplied and redistributed, creating a tension between individual incentive and collective benefit [6]. Classical analyses assume a homogeneous population, but real groups are heterogeneous: players differ in their productivity, their contributions, and how sensitively they respond to payoff differences [5].

A natural learning rule for such settings is *introspection dynamics*, introduced in [1]: at each time step a player compares their current payoff to the payoff they *would have received* had they played the alternative action (keeping all other players fixed), and switches with a probability governed by this difference. Because the comparison is self-referential rather than imitative, introspection applies to arbitrary asymmetric games where role-model imitation is unavailable or unnatural.

In [2] the authors extended introspection dynamics to multiplayer asymmetric games and derived a formula for the stationary distribution that can be evaluated numerically for any game. In their general framework the transition rate for player i switching depends on the actions of all other players through the payoff function, so the stationary distribution must be computed by solving a linear system of dimension 2^N; no closed form is available in general.

In this paper we show that whenever the payoff difference Δf_i a player evaluates when considering a switch is independent of all other players' current actions, a property we call *state-independence*, the introspection chain decomposes exactly as a product of N independent two-state chains (one per player), the stationary distribution is a product measure, and the cooperation probability p_C^{intro} admits an explicit closed form scaling as $O(N)$ rather than $O(2^N)$. We then show that the heterogeneous public goods game (PGG) satisfies state-independence as a consequence of the *linearity* of its payoff structure, yielding the exact formula (8).

The paper is organised as follows. Section 2 introduces the general framework and the state-independence condition, noting that whilst the donation-form Prisoner's Dilemma is state-independent, the Stag Hunt and the classical (RPST) Prisoner's Dilemma are not. Section 3 proves, for any state-independent game, that individual cooperation probabilities are given by (4) and the stationary distribution is a product measure. Section 4 establishes state-independence for the PGG and derives the exact cooperation formula. Section 5 records several structural corollaries. Section 6 concludes.

2 Setup and notation

Consider N ordered players, each choosing from two actions $\{C, D\}$ coded as $\{1, 0\}$. The state space is $S = \{0,1\}^N$ with $|S| = 2^N$. Each player i has a payoff function $f_i : S \to \mathbb{R}$ and player-specific selection intensity $\beta_i > 0$. Writing $\mathbf{a}_{i\to\bar{a}_i}$ for the state with player i's action flipped to $\bar{a}_i = 1 - a_i$, the *payoff difference* for player i at state $\mathbf{a}$ is

$$\Delta f_i(\mathbf{a}) := f_i(\mathbf{a}) - f_i(\mathbf{a}_{i\to\bar{a}_i}).$$

A payoff function f_i is *state-independent* if $\Delta f_i(\mathbf{a})$ depends only on a_i, not on $\mathbf{a}_{-i}$, the actions of all individuals other than i. When this holds there is a constant $\delta_i \in \mathbb{R}$ such that

$$\Delta f_i(\mathbf{a}) = (1 - 2a_i)\,\delta_i, \tag{1}$$

where $\delta_i = \Delta f_i\big|_{a_i=0}$ is the payoff difference when player i defects. When $a_i = 0$, (1) follows directly from the definition of δ_i; when $a_i = 1$, $\Delta f_i\big|_{a_i=1} = -\Delta f_i\big|_{a_i=0} = -\delta_i$, giving (1).

Note that not all games lead to *state-independent* payoff functions. For example in [1] two of the 2 player, 2 strategy games considered to illustrate introspection dynamics are: the Prisoner's Dilemma (2 - M_1) and the Stag-hunt (2 - M_2) for $0 < c_1, c_2 < b$.

$$M_1 = \begin{pmatrix} b - c_1, b - c_2 & -c_1, b \\ b, -c_2 & 0, 0 \end{pmatrix} \qquad M_2 = \begin{pmatrix} b - c_1, b - c_2 & -c_1, 0 \\ 0, -c_2 & 0, 0 \end{pmatrix} \tag{2}$$

For M_1: $\delta_1 = c_1$ and $\delta_2 = c_2$ however for M_2, $\Delta f_1((1,1)) = b - c_1$ but $\Delta f_1((1,0)) = -c_1$. The Prisoner's Dilemma (as defined by M_1) is *state-independent* but the Stag-hunt game (as defined by M_2) is not. Further games considered in [1] are the Volunteer's dilemma and the Volunteer's timing dilemma (a generalisation with more than 2 strategies) neither of which are *state-independent*.

More generally, the classical Prisoner's Dilemma is parameterised by the reward R, temptation T, sucker payoff S, and punishment P with $T > R > P > S$ and the social-efficiency condition

$2R > T + S$. State-independence for player 1 requires the *equality*

$$T - R \;=\; P - S \quad (\text{equivalently } P + R = S + T),$$

that is, the gain from defecting against a cooperator must equal the gain from defecting against a defector. The PD axioms are strict inequalities and do not force this equality; a generic RPST Prisoner's Dilemma therefore fails it. For instance, $T = 5$, $R = 3$, $P = 1$, $S = 0$ satisfies $T > R > P > S$ and $2R = 6 > 5 = T + S$, yet $T - R = 2 \neq 1 = P - S$. The donation form M_1 above is the exceptional case where the equality holds by construction (both differentials equal c).

To define introspection dynamics we introduce the following two quantities:

- **Fermi function.** Each player i has a personal Fermi function $\phi_i(x) = (1 + e^{\beta_i x})^{-1}$. Since $\beta_i > 0$, the exponential $e^{\beta_i x}$ is strictly increasing in x, so ϕ_i is strictly decreasing with $\phi_i(0) = \frac{1}{2}$ and $\phi_i(x) + \phi_i(-x) = 1$.
- **Mutation.** Each player i has mutation probabilities $\mu_{i0}, \mu_{i1} > 0$ with $\mu_{i0} + \mu_{i1} < 1$, where μ_{i0} is the probability of mutating to cooperation and μ_{i1} is the probability of mutating to defection, regardless of the intended choice of the dynamic.

Introspection dynamics. At each step, one player i is selected uniformly at random. Player i considers switching from action a_i to the alternative $\bar{a}_i = 1 - a_i$. The transition probability for the switch is

$$T^{\text{intro}}_{\mathbf{a},\, \mathbf{a}_{i \to \bar{a}_i}} = \frac{1}{N}\left[(1 - \mu_{i0} - \mu_{i1})\phi_i(\Delta f_i(\mathbf{a})) + \mu_{i,\bar{a}_i}\right], \tag{3}$$

where $\mu_{i,\bar{a}_i}$ denotes μ_{i0} when $\bar{a}_i = C$ and μ_{i1} when $\bar{a}_i = D$. Since ϕ_i is strictly decreasing, a larger payoff gain from keeping the current action yields a smaller switching probability.

Definition 1 (Cooperation probability)**.** *Given an ergodic chain on S with stationary distribution π, the* cooperation probability *is $p_C = \pi \cdot s$, where $s_{\mathbf{a}} = \frac{1}{N}\sum_i a_i$ is the fraction of cooperators in state $\mathbf{a}$.*

3 Individual cooperation probabilities

For a state-independent game, $\Delta f_i(\mathbf{a})$ depends only on a_i, not on $\mathbf{a}_{-i}$. We use this to compute each player's long-run cooperation probability and the stationary distribution directly from the two-state update rule.

Theorem 1 (Individual cooperation probabilities)**.** *For any state-independent game with constants δ_i, under introspection dynamics with player-specific selection intensity β_i and mutation probabilities $\mu_{i0}, \mu_{i1} > 0$ with $\mu_{i0} + \mu_{i1} < 1$, the long-run probability that player i cooperates is*

$$p_i = \phi_i(\delta_i)(1 - \mu_{i0} - \mu_{i1}) + \mu_{i0}. \tag{4}$$

Proof. By state-independence, $\Delta f_i(\mathbf{a}) = (1 - 2a_i)\delta_i$ depends only on a_i. When player i is selected, the probability their next action is C equals p_i regardless of their current action: if $a_i = D$, the switch probability is $p_i = \phi_i(\delta_i)(1 - \mu_{i0} - \mu_{i1}) + \mu_{i0}$; if $a_i = C$, the probability of remaining at C is $1 - [\phi_i(-\delta_i)(1 - \mu_{i0} - \mu_{i1}) + \mu_{i1}]$, using $\phi_i(x) + \phi_i(-x) = 1$ this gives: $1 - [(1 - \phi_i(\delta_i))(1 -$

$\mu_{i0} - \mu_{i1}) + \mu_{i1}] = p_i$. Since each selection of player i places them in state C with probability p_i independently of their current state, the long-run cooperation probability of player i is p_i.

□

Theorem 2 (Stationary distribution). *For any state-independent game with constants δ_i, the stationary distribution of the introspection chain on $S = \{0,1\}^N$ is the product measure*

$$\pi_{\mathbf{a}} = \prod_{i=1}^{N} p_i^{a_i}(1-p_i)^{1-a_i}, \qquad \mathbf{a} \in S, \tag{5}$$

where p_i is given by (4)*.*

Proof. By Theorem 1, each selection of player i places them in state C with probability p_i independently of every other player's action. In stationarity the players' actions are therefore independent, with player i cooperating with probability p_i and defecting with probability $1-p_i$. The probability of state $\mathbf{a}$ is the product of these marginals, giving (5).

□

Remark 3. *For the Prisoner's Dilemma M_1 (2) of [1] the payoff $f_1(\mathbf{a}) = -c_1a_1 + ba_2$ is separable with $\delta_1 = c_1$, and similarly $\delta_2 = c_2$. With $b = 1$, $c_1 = 0.6$, $c_2 = 0.1$, $\beta = 5$, and $\mu_{ij} = 0$, Theorems 1 and 2 give*

$$p_1 = \frac{1}{1+e^3} \approx 0.047, \qquad p_2 = \frac{1}{1+e^{0.5}} \approx 0.378,$$

and the stationary distribution over $\{DD, DC, CD, CC\}$ is the product measure

$$\pi_{DD} \approx 0.593, \quad \pi_{DC} \approx 0.360, \quad \pi_{CD} \approx 0.030, \quad \pi_{CC} \approx 0.018.$$

4 The heterogeneous public goods game

The *heterogeneous public goods game* is played by N players with player-specific contributions $\alpha_1, \ldots, \alpha_N > 0$ and public goods multipliers $r_1, \ldots, r_N > 1$. The payoff to player i in state $\mathbf{a} = (a_1, \ldots, a_N)$ is

$$f_i(\mathbf{a}) = \frac{r_i}{N} \sum_{j=1}^{N} \alpha_j a_j - \alpha_i a_i. \tag{6}$$

We write $C(\mathbf{a}) = \sum_j \alpha_j a_j$ for the total contribution.

Lemma 4 (State-independence of the PGG). *The payoff* (6) *is state-independent for every player i, with*

$$\delta_i = \alpha_i\left(1 - \frac{r_i}{N}\right), \tag{7}$$

i.e. $\Delta f_i(\mathbf{a}) = (1-2a_i)\,\alpha_i(1 - r_i/N)$, independent of $\mathbf{a}_{-i}$.

Proof. When player i switches from a_i to $\bar{a}_i = 1 - a_i$, only their own contribution changes, so the total contribution becomes $C(\mathbf{a}_{i\to\bar{a}_i}) = C(\mathbf{a}) + \alpha_i(1-2a_i)$. Expanding both payoffs:

$$\begin{aligned} f_i(\mathbf{a}) &= \frac{r_i}{N} C(\mathbf{a}) - \alpha_i a_i, \\ f_i(\mathbf{a}_{i\to\bar{a}_i}) &= \frac{r_i}{N}\big[C(\mathbf{a}) + \alpha_i(1-2a_i)\big] - \alpha_i(1-a_i). \end{aligned}$$

Subtracting:

$$\begin{aligned}\Delta f_i &= -\frac{r_i}{N}\alpha_i(1-2a_i) + \alpha_i(1-2a_i)\\ &= \alpha_i(1-2a_i)\Big(1-\frac{r_i}{N}\Big).\end{aligned}$$

The term $C(\mathbf{a})$ cancels entirely, confirming state-independence with $\delta_i = \alpha_i(1 - r_i/N)$. □

Corollary 5 (Exact cooperation probability)**.** *For the heterogeneous public goods game* (6)*, the long-run cooperation probability is*

$$\boxed{p_C^{\mathrm{intro}} = \frac{1}{N}\sum_{i=1}^{N}\left[\frac{1-\mu_{i0}-\mu_{i1}}{1+e^{\beta_i\alpha_i(1-r_i/N)}} + \mu_{i0}\right].} \tag{8}$$

Proof. By Lemma 4, $\delta_i = \alpha_i(1 - r_i/N)$. By Theorem 1: $p_C^{\mathrm{intro}} = \frac{1}{N}\sum_{i=1}^{N} p_i$. Substituting $\phi_i(\delta_i) = (1 + e^{\beta_i\alpha_i(1-r_i/N)})^{-1}$ into (4) gives (8).

□

Figure 1 validates Corollary 5 against exact values and simulation across group sizes. Table 1 verifies Theorem 2: for $N = 3$ with per-player multipliers $r_1 = 1$, $r_2 = 3 = N$, $r_3 = 9$, uniform contributions $\alpha_i = 1$, $\beta_i = 2$, and $\mu_{i0} = \mu_{i1} = 0.1$, the individual cooperation probabilities are $p_1 \approx 0.267$, $p_2 = 0.500$, $p_3 \approx 0.886$, and the formula and exact values of $\pi_{\mathbf{a}}$ agree to five decimal places for all eight states.

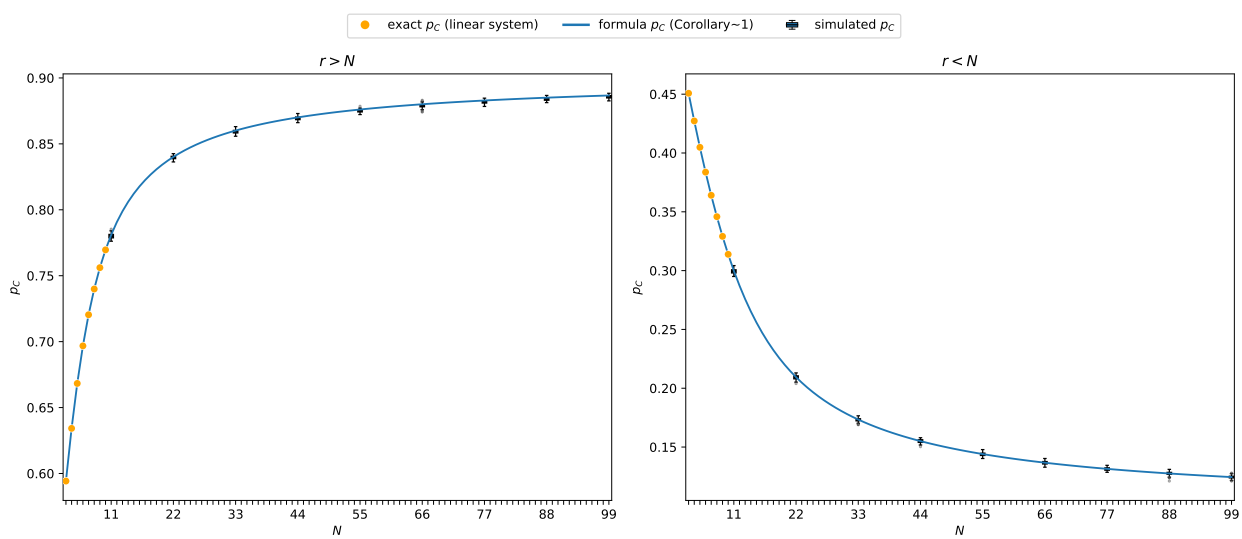

Figure 1: Validation of formula (8) on a heterogeneous group with $\alpha_i = i$, $\beta = 0.5$, $\mu_{i0} = \mu_{i1} = 0.1$. *Left:* $r = 2N$ (so $r > N$ for all N). *Right:* $r = N/2$ (so $r < N$ for all N). Orange dots: exact p_C from the full 2^N linear system (feasible only for small N). Blue curve: formula (8) (Corollary 5). Blue boxes: distribution of p_C across 99 independent runs of 10^5 simulation steps each (outliers shown). The negligible variability across runs reflects the large number of simulation steps per run. All values obtained using [4].

State $\mathbf{a}$	Formula $\pi_{\mathbf{a}}$	Exact $\pi_{\mathbf{a}}$
DDD	0.04193	0.04193
DDC	0.32463	0.32462
DCD	0.04193	0.04193
DCC	0.32463	0.32462
CDD	0.01526	0.01527
CDC	0.11818	0.11818
CCD	0.01526	0.01527
CCC	0.11818	0.11818

Table 1: Stationary-distribution probabilities $\pi_{\mathbf{a}}$ for all eight states $\mathbf{a} \in \{D, C\}^3$ with $N = 3$, $\alpha_i = 1$, $\beta = 2$, $\mu_{i0} = \mu_{i1} = 0.1$, and per-player multipliers $r_1 = 1 < N = r_2 = 3 < r_3 = 9$. Formula: product measure (5) (Theorem 2); Exact: values from the 2^N linear system obtained using [4]. Both columns to five decimal places.

5 Structural consequences

The formulas (4) and (8) allow immediate structural conclusions about individual and aggregate cooperation.

Corollary 6 (Player-specific cooperation thresholds)**.** *Player i cooperates with probability greater than $\frac{1}{2}$ if and only if*

$$\phi_i(\delta_i) > \frac{\frac{1}{2} - \mu_{i0}}{1 - \mu_{i0} - \mu_{i1}}.$$

For the PGG, $\delta_i = \alpha_i(1 - r_i/N)$; when $\mu_{i0} = \mu_{i1}$ this reduces to $r_i > N$. A sufficient condition for $p_C^{\text{intro}} > \frac{1}{2}$ is that the inequality holds for every i.

Proof. Since $1 - \mu_{i0} - \mu_{i1} > 0$:

$$p_i > \tfrac{1}{2} \iff (1 - \mu_{i0} - \mu_{i1})\phi_i(\delta_i) > \tfrac{1}{2} - \mu_{i0} \iff \phi_i(\delta_i) > \frac{\frac{1}{2} - \mu_{i0}}{1 - \mu_{i0} - \mu_{i1}}.$$

When $\mu_{i0} = \mu_{i1}$, the right-hand side equals $\frac{1}{2}$, and since ϕ_i is strictly decreasing with $\phi_i(0) = \frac{1}{2}$, this is equivalent to $\delta_i < 0$. For the PGG, $\delta_i = \alpha_i(1 - r_i/N) < 0$ iff $r_i > N$. The claim about p_C^{intro} follows by averaging.

□

With symmetric mutation, a player cooperates the majority of the time precisely when their personal multiplier on the public good exceeds the group size: that is, when they extract more from the pool per unit contributed than they put in. The top centre panel of Figure 2 illustrates this, with curves at $r > N$ lying above $p_i = \frac{1}{2}$ and curves at $r < N$ lying below it.

Corollary 7 (Neutral drift)**.** *If $\beta_i = 0$ for all i, then ϕ_i is constant and the payoff parameters δ_i have no effect on p_C; cooperation is determined entirely by mutation:*

$$p_i = \frac{1 + \mu_{i0} - \mu_{i1}}{2}, \qquad p_C^{\text{intro}} = \frac{1}{2} + \frac{1}{2N}\sum_{i=1}^{N}(\mu_{i0} - \mu_{i1}).$$

In particular, if $\beta_i = 0$ for all i, $p_C^{\text{intro}} = \frac{1}{2}$ if and only if $\sum_i(\mu_{i0} - \mu_{i1}) = 0$.

Proof. When $\beta_i = 0$, the Fermi function satisfies $\phi_i(x) = \frac{1}{2}$ for all x. Substituting into (4):

$$p_i = \frac{1 - \mu_{i0} - \mu_{i1}}{2} + \mu_{i0} = \frac{1}{2} + \frac{\mu_{i0} - \mu_{i1}}{2} = \frac{1 + \mu_{i0} - \mu_{i1}}{2}. \qquad \square$$

Players who are insensitive to payoff differences ignore the underlying game; their long-run behaviour is set entirely by their mutation bias, and group-level cooperation reduces to the average asymmetry between mutation rates. The top right panel of Figure 2 illustrates this for symmetric mutation: at $\beta_i = 0$, $p_i = \frac{1}{2}$ for every choice of α_i and r_i. The top left panel shows the same from a different angle: all three curves converge to $p_i = \frac{1}{2}$ at $\beta_i = 0$ before diverging as sensitivity increases, with the direction of divergence set by r_i relative to N.

Corollary 8 (Role of heterogeneity)**.** *For the PGG, for player i with $r_i \neq N$:*

(i) $\dfrac{\partial p_i}{\partial \alpha_i}$ *is positive when $r_i > N$ and negative when $r_i < N$.*

(ii) $\dfrac{\partial p_i}{\partial \beta_i}$ *is positive when $r_i > N$ and negative when $r_i < N$.*

Proof. Write $p_i = (1 - \mu_{i0} - \mu_{i1})\phi_i(\alpha_i(1 - r_i/N)) + \mu_{i0}$.

Part (i). Differentiating with respect to α_i:

$$\frac{\partial p_i}{\partial \alpha_i} = (1 - \mu_{i0} - \mu_{i1})(1 - r_i/N)\, \phi_i'(\alpha_i(1 - r_i/N)).$$

Since $\phi_i'(x) < 0$ for all x and $1 - \mu_{i0} - \mu_{i1} > 0$, the sign equals $-\mathrm{sgn}(1 - r_i/N) = \mathrm{sgn}(r_i/N - 1)$, which is positive if and only if $r_i > N$.

Part (ii). Write $g(t) = (1 + e^t)^{-1}$, so $\phi_i(x) = g(\beta_i x)$. Differentiating with respect to β_i:

$$\frac{\partial p_i}{\partial \beta_i} = (1 - \mu_{i0} - \mu_{i1})\, \alpha_i(1 - r_i/N)\, g'(\beta_i \alpha_i(1 - r_i/N)).$$

Since $g' < 0$ and $\alpha_i > 0$, the sign equals $-\mathrm{sgn}(1 - r_i/N) = \mathrm{sgn}(r_i/N - 1)$, which is positive if and only if $r_i > N$. $\square$

For a player who benefits from the public good ($r_i > N$), contributing more (larger α_i) or responding more strongly to payoff differences (larger β_i) raises their cooperation rate; the same changes lower it for a player who does not benefit ($r_i < N$). Heterogeneity in any one of these parameters therefore translates into heterogeneity in cooperation, with the sign of the effect set by r_i relative to N. The bottom centre and right panels of Figure 2 show this directly: the heatmaps of $\partial p_i/\partial \alpha_i$ and $\partial p_i/\partial \beta_i$ change sign exactly at $r_i = N$. The bottom left panel shows that $\partial p_i/\partial \mu$ obeys the same rule: raising the mutation rate pulls p_i towards $\frac{1}{2}$, lowering cooperation when $r_i > N$ and raising it when $r_i < N$.

Remark 9. *Corollaries 6–8 show that each player's long-run cooperation is fully determined by their individual tuple $(\alpha_i, \beta_i, r_i, \mu_i)$ and the group size N. Players with $r_i > N$ cooperate more than half the time, and increasing α_i or β_i raises their cooperation probability further; players with $r_i < N$ show the opposite sensitivity. The parameters of other players have no effect on player i's marginal cooperation probability.*

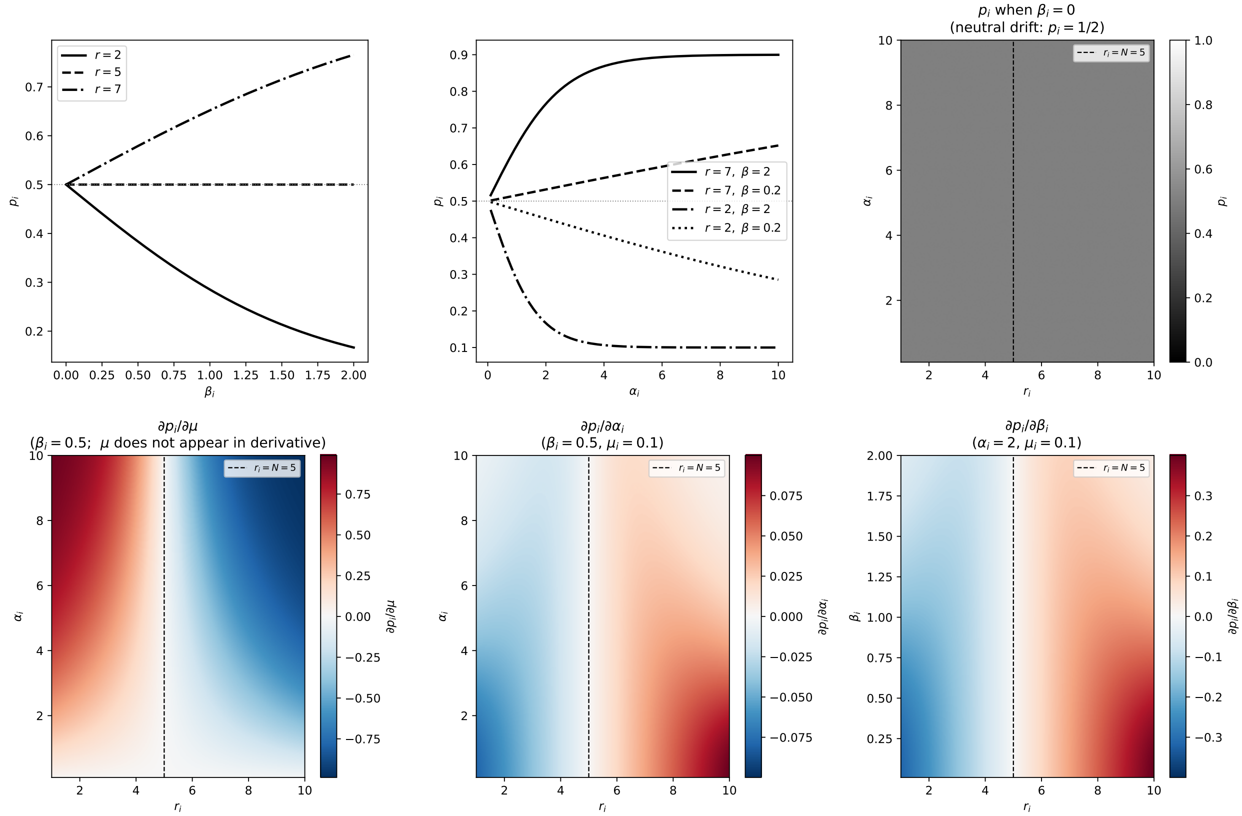

Figure 2: Structural properties of p_i for a single player with $N = 5$, $\mu_{i0} = \mu_{i1} = 0.1$. *Top left:* p_i as a function of β_i ($\alpha_i = 2$) for $r < N$, $r = N$, and $r > N$; all curves pass through $p_i = \frac{1}{2}$ at $\beta_i = 0$ (Corollary 7). *Top centre:* p_i as a function of α_i for two values of r and β_i; curves above (below) $\frac{1}{2}$ correspond to $r > N$ ($r < N$). *Top right:* $p_i(\alpha_i, r_i)$ at $\beta_i = 0$, confirming $p_i = \frac{1}{2}$ everywhere (Corollary 7); the dashed line marks $r_i = N$. *Bottom row:* Heatmaps of the partial derivatives $\partial p_i/\partial \mu_i$, $\partial p_i/\partial \alpha_i$, and $\partial p_i/\partial \beta_i$ over the (r_i, α_i) or (r_i, β_i) plane ($\beta_i = 0.5$ or $\alpha_i = 2$ as fixed parameter). Red indicates a positive derivative; blue indicates negative. Each derivative changes sign exactly at $r_i = N$ (dashed line), consistent with Corollary 8.

6 Conclusion

We have proved that introspection dynamics on any state-independent game is exactly solvable. When Δf_i depends only on a_i, each selection of player i places them in state C with probability p_i regardless of current state (Theorem 1), and the stationary distribution is a product measure (Theorem 2). For the heterogeneous public goods game, the linear payoff structure implies state-independence (Lemma 4), and the long-run cooperation probability has a closed form with N terms rather than 2^N (Corollary 5).

The approach does not extend to games that are not state-independent. When $\Delta f_i(\mathbf{a})$ depends on $\mathbf{a}_{-i}$, the per-player chains are coupled and the stationary distribution is no longer a product measure; the full 2^N linear system must be solved in general. Games such as the Stag Hunt (M_2 of (2)) contain payoff cross terms $a_i a_{-i}$ that cannot be eliminated by rescaling, so no reformulation brings them within the scope of the present results.

The source code, figures, and data for this paper are under version control and available at [3], in line with best practices for research software [7].

7 Acknowledgements

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